

Product Requirements Document (PRD)

AI-Based Training Centre Compliance & Monitoring System

Version: 1.0 **Product Type:** Government Compliance Intelligence Platform
Primary Users: Government monitoring authorities, monitoring officers, training-centre operators

1. Executive Summary

The product is an AI-powered, privacy-preserving video analytics platform for monitoring government-funded training centres. It uses periodic camera/CCTV snapshots rather than continuous high-quality video transmission to estimate physical attendance, inspect approved infrastructure and seating capacity, assess workshop availability, and detect apparent equipment-condition issues.

The system compares centre-reported information with AI-observed evidence and generates actionable, evidence-backed compliance alerts. AI should flag anomalies rather than automatically declare a centre fraudulent; a monitoring officer reviews the evidence and can accept, reject, or request physical inspection.

2. Problem Statement

Government monitoring teams may receive reported attendance and infrastructure information from training centres but may not have an efficient way to continuously verify whether the reported conditions match what is physically present.

Manual CCTV review is time-consuming and continuous video upload is bandwidth-intensive. The product addresses this by extracting periodic frames at the edge, processing them with computer vision, and sending only necessary metadata, alerts, confidence information, and selected evidence to the central dashboard.

3. Product Vision

Convert existing CCTV/camera recordings into actionable compliance intelligence while minimizing bandwidth use and unnecessary collection of personally identifying information.

4. Goals & Objectives

Verify reported attendance against AI-estimated physical presence.

Compare approved infrastructure inventory against visually detected infrastructure.

Estimate seating availability and workshop presence/availability.

Detect visually apparent equipment damage or unusable conditions where detectable.

Generate evidence-backed alerts with confidence and severity.

Provide centre-wise compliance scores and historical trends.

Enable human verification before formal compliance action.

Use privacy-preserving counting rather than identity-based face recognition for normal monitoring.

5. Target Users

Monitoring Authority: Oversees many centres through a central dashboard.

Monitoring Officer: Reviews alerts, evidence, history, and takes verification actions.

Training Centre Operator: Submits official attendance and maintains approved infrastructure information.

System Administrator: Manages centres, users, approved inventory, camera configuration, thresholds, and system health.

6. Core User Stories

As a monitoring officer, I want to see all centres and their compliance status so I can prioritize attention.

As a monitoring officer, I want to compare reported attendance with AI-estimated presence so discrepancies are visible.

As an authority, I want approved infrastructure compared with detected infrastructure so shortages are automatically highlighted.

As an officer, I want every alert to include supporting snapshot evidence and confidence so I can make an informed decision.

As an officer, I want to accept, reject, or request physical inspection of an alert.

As an authority, I want historical compliance trends so I can identify centres with repeated deterioration.

As a centre operator, I want to submit official attendance before monitoring begins.

As a system administrator, I want to configure approved inventory, seating capacity, workshop requirements, and monitoring thresholds.

7. Functional Requirements

FR-01 Attendance Submission: Centre submits/records official attendance at the beginning of each training day.

FR-02 Snapshot Extraction: Edge system extracts periodic frames from CCTV/camera recordings at a configurable interval, e.g. 1–5 minutes.

FR-03 Person Counting: Computer vision estimates the number of people physically present without identifying individuals.

FR-04 Attendance Comparison: System compares reported attendance with AI-estimated presence and calculates discrepancy.

FR-05 Infrastructure Inventory: Store approved/sanctioned inventory for each centre, including type and approved quantity.

FR-06 Infrastructure Detection: Detect relevant equipment such as workbenches, machines, chairs, computers and other configured categories.

FR-07 Infrastructure Comparison: Compare approved quantity with detected quantity and identify shortages, additional/unapproved items where relevant, and potential damage.

FR-08 Seating Verification: Compare approved seating capacity with visually detected/available seating.

FR-09 Workshop Check: Determine whether required workshop area/equipment is visually detectable and flag missing requirements.

FR-10 Apparent Operability: Detect visual indicators such as damage, missing components, visible unusability, or powered-on/off status where detectable. The system must not claim definitive mechanical/electrical functionality.

FR-11 Privacy Processing: Apply face blurring/anonymisation where people appear in evidence imagery and avoid identity recognition for normal compliance monitoring.

FR-12 Compliance Engine: Evaluate attendance, infrastructure, seating, workshop, and historical signals against configured rules.

FR-13 Alert Engine: Generate alerts with category, severity, discrepancy, confidence, timestamp, centre, and evidence reference.

FR-14 Evidence: Link each alert to the relevant snapshot and AI result/comparison.

FR-15 Dashboard: Display all centres with attendance status, infrastructure status, alert count, and overall status.

FR-16 Centre Details: Show centre ID/name, location, reported/AI attendance, discrepancy, seating, infrastructure, workshop status, current/historical alerts, evidence, and compliance percentage.

FR-17 Human Verification: Officer can Accept, Reject, or Request Physical Inspection for an alert.

FR-18 Compliance Score: Calculate a configurable centre score from attendance, infrastructure, seating, workshop, historical alerts, and previous response/SLA signals.

FR-19 Historical Trends: Provide Today, This Week, This Month, and historical views.

FR-20 Performance Evaluation: Maintain ground-truth vs AI prediction records for prototype evaluation and calculate measured accuracy/false-positive/false-negative metrics.

8. Alert Severity

Low: Small discrepancy or low-confidence observation.

Medium: Repeated discrepancy or moderate infrastructure shortage.

Critical: Large attendance mismatch, major infrastructure shortage, or repeated compliance failure.

9. Central Monitoring Dashboard

The main dashboard should provide an all-centres table with centre, attendance status, infrastructure status, alert count, and overall compliance status. Selecting a centre opens its detailed compliance view.

Example statuses: Compliant, Attention, Critical. The dashboard should prioritize actionable alerts rather than forcing officers to manually inspect hours of CCTV.

10. Alert Evidence Workflow

Every AI-generated alert follows: Snapshot → AI result → comparison → alert → human review.

Evidence should show the reported value, observed value, difference, confidence, timestamp, and source snapshot. The wording must remain probabilistic and evidence-based rather than asserting fraud.

11. Privacy & Responsible AI Requirements

Normal monitoring must focus on counting people rather than identifying individual trainees.

Use person detection → counting → face blurring/anonymisation → metadata rather than face recognition → identity → attendance.

Evidence images should be minimized and access-controlled. Retention periods should be configurable according to the applicable government policy.

AI outputs must be treated as observations/estimates. Human review is required before adverse compliance action.

Camera angle, occlusion, poor lighting, movement, temporary equipment relocation, and camera/network failures must be considered as sources of error.

12. Low-Bandwidth Architecture

Camera/CCTV Recording → Local Edge Computer → Periodic Frame Extraction → AI Computer Vision Analysis → Privacy Processing → Compliance Engine → Alert Engine → Central Dashboard.

Only required metadata, alerts, confidence values, and selected evidence should be transmitted to the cloud/central platform where possible. Continuous high-quality video streaming to the central server is not required for the core workflow.

13. High-Level Technical Architecture

Edge Layer: Camera/CCTV input, frame extraction, local inference, privacy processing, local buffering.

AI/CV Layer: Person counting, object/infrastructure detection, seating detection, workshop detection, apparent-condition detection.

Compliance Layer: Rules engine for reported-vs-observed comparisons, thresholds, severity, scoring, and trend calculations.

Backend Layer: APIs, authentication/authorization, centre/inventory/attendance/alert/evidence data management.

Storage Layer: Relational database for structured compliance data and object storage for selected evidence snapshots.

Dashboard Layer: Web application for all-centres monitoring, centre details, alerts, evidence review, analytics, and verification actions.

14. Suggested Data Model

Centre: centre_id, name, location, status, approved_seating_capacity, workshop_requirement.

AttendanceRecord: attendance_id, centre_id, date, reported_count, submitted_at, submitted_by.

DetectionSnapshot: snapshot_id, centre_id, captured_at, source_camera, people_count, confidence, evidence_uri.

InfrastructureItem: item_id, centre_id, category, approved_quantity, required_condition.

DetectionResult: result_id, snapshot_id, category, detected_quantity, condition, confidence.

ComplianceResult: compliance_id, centre_id, date/time, category, expected_value, observed_value, discrepancy, confidence.

Alert: alert_id, centre_id, category, severity, status, discrepancy, confidence, evidence_id, created_at.

Verification: verification_id, alert_id, officer_id, decision, remarks, reviewed_at.

CentreScore: centre_id, period, attendance_score, infrastructure_score, seating_score, workshop_score, historical_score, overall_score.

15. API Requirements

POST /centres/{id}/attendance — submit official attendance.

GET /centres — list centres with compliance summary.

GET /centres/{id} — retrieve centre details.

GET /centres/{id}/alerts — retrieve alerts.

GET /alerts/{id}/evidence — retrieve authorized evidence metadata/snapshot.

POST /alerts/{id}/verification — accept, reject, or request physical inspection.

GET /centres/{id}/trends — historical compliance trends.

GET /centres/{id}/infrastructure — approved vs detected infrastructure.

GET /analytics/performance — measured AI evaluation metrics.

16. UI/UX Requirements

Primary screens: Login, All Centres Dashboard, Centre Details, Alerts, Alert Evidence Review, Historical Analytics, Infrastructure Inventory, Attendance Submission, System/Camera Health, and AI Performance Evaluation.

Use clear status indicators, compact summary cards, tables for centre-level comparison, charts for trends, and an evidence-first alert review experience.

Critical alerts must be visually distinguishable but should not imply guilt before human verification.

17. Non-Functional Requirements

Bandwidth: Core workflow should minimize continuous video transmission.

Privacy: Minimize personally identifying data and restrict evidence access.

Reliability: Edge system should buffer events during temporary network outages and sync when connectivity returns.

Scalability: Architecture should support monitoring many centres without requiring manual CCTV review.

Auditability: Record alert creation, evidence reference, and human verification actions.

Security: Role-based access, encrypted transport, secure credential handling, and controlled evidence access.

Observability: Monitor camera health, edge device health, inference failures, sync failures, and API health.

18. Edge Cases & Error Handling

Poor lighting or occlusion may reduce detection confidence.

Camera offline or disconnected → mark camera health issue and avoid interpreting missing data as zero attendance.

Network unavailable → queue results locally and synchronize later.

Temporary equipment movement → use repeated observations and human review.

Duplicate snapshots → deduplicate using capture metadata.

Low-confidence detection → downgrade severity or require review rather than making a strong claim.

Multiple camera views → avoid double-counting people where the prototype supports multiple cameras.

19. AI Evaluation Strategy

Create a labelled test dataset covering different attendance levels, camera angles, lighting conditions, occlusion, seating arrangements, and infrastructure conditions.

Compare ground truth with model predictions. Calculate measured attendance/infrastructure detection accuracy, precision/recall where applicable, false-positive rate, and false-negative rate.

Do not use illustrative accuracy numbers in the final SIH presentation unless they were actually measured on the prototype dataset.

20. Compliance Score

Provide a 0–100 centre compliance score using configurable weights for attendance, infrastructure, seating, workshop compliance, historical alerts, and response to previous alerts. The exact weighting should be configurable rather than hard-coded until stakeholder validation.

21. Testing Strategy

Unit: counting/comparison logic, scoring, severity rules, API validation.

AI: labelled dataset evaluation across controlled scenarios.

Integration: camera/edge → backend → dashboard flow.

Privacy: verify anonymisation and access restrictions.

Reliability: network outage, edge buffering, recovery, duplicate frames.

UX: alert review, evidence display, filters, historical views.

Security: authorization, evidence access, input validation, audit logs.

22. KPIs & Analytics

Attendance discrepancy rate per centre.

Infrastructure shortage rate.

Workshop compliance rate.

Average alert confidence.

Alert resolution time.

Percentage of alerts reviewed by officers.

Repeat-alert rate by centre.

Centre compliance score trend.

Measured AI false-positive and false-negative rates.

Camera/edge uptime and data synchronization success rate.

23. Risks & Mitigations

False positives: use confidence thresholds and human verification.

Privacy concerns: avoid identity recognition and anonymise evidence.

Bandwidth limitations: edge processing and periodic snapshots.

Camera quality: camera-health checks and confidence-aware alerts.

AI cannot prove internal machine functionality: label outputs as apparent operability and use physical/sensor verification where needed.

Evidence misuse: role-based access, audit logs, and configurable retention.

24. MVP Scope

Must have: centre registry, attendance submission, periodic snapshots, person counting, approved infrastructure inventory, basic infrastructure/seating detection, discrepancy engine, alert engine, privacy processing, central dashboard, evidence snapshots, human verification, and basic compliance score.

Should have: historical trends, workshop checks, apparent operability, camera/edge health, AI evaluation dashboard.

Could have: advanced multi-camera fusion, automated inspection scheduling, richer analytics, predictive compliance risk scoring.

25. Development Milestones

Phase 1: Centre/inventory/attendance data model and dashboard shell.

Phase 2: Camera simulation + snapshot extraction + edge pipeline.

Phase 3: Person counting and infrastructure detection prototype.

Phase 4: Compliance comparison, severity, alerts, evidence.

Phase 5: Privacy processing and human verification workflow.

Phase 6: Historical analytics, score, evaluation dataset, testing.

Phase 7: SIH demo hardening, documentation, deployment and presentation.

26. Final Product Positioning

An AI-powered, privacy-preserving, low-bandwidth Training Centre Compliance Intelligence Platform that converts existing CCTV recordings into actionable compliance insights by monitoring attendance, seating capacity, infrastructure availability, workshop facilities, and apparent equipment condition, while providing evidence-backed alerts and centre-wise compliance analytics to government monitoring authorities.

27. Important Product Language

Do not position the system as “AI will tell whether attendance is fake or genuine.”

Preferred wording: **“AI will estimate physical presence from camera snapshots and automatically identify discrepancies between reported attendance and observed presence, generating evidence-backed compliance alerts for human verification.”**